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SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM - 2

Programme: B.E-EEE
Course Code : S11BLH22
Course Name: Python Programming
Batch : 2023-2027

Max. Marks: 50
Time: 2 Hours
Sem : 3
Date: 26.09.2024

Part-A Answer ANY FIVE questions (5*10=50)

Note: Write Python Coding and Expected Output for all the questions

Q.No	Questions	CO
1.	Given the following dataset [25,30,15,57,66], write a python program to calculate the mean,variance and standard deviation. Here are some steps to calculate the standard deviation. 1. Find the mean by adding all the numbers and dividing by the total number of data points 2. Subtract each data point from the mean 3. Square each deviation from mean 4. Add up all the squared value 5. Divide the sum by total number of data point to get variance 6. Take the square root of the result of step 5 to get the standard deviation	CO3
2.	a) Given the data set [7,5,8,1,7,4], write a python program to find and print all pairs of numbers from the list that sum up to 12. **Example** [4,1,2,3,4] pairs of numbers which gives the sum as 5 (4,1) (2,3) (3,2) (7,5) (5,7) b) Write a python script to generate and print a dictionary that contains a number between 1 and n in the form (x,x³) (8,4) **Example** (x,x²) {1:1,2:4,3:9,4:16,5:25} 25 64	CO3
3.	Write a python program that prompts the use to input a paragraph. Store the input in a list, with each sentence or word as an individual element and finally display those sentences as a paragraph.	CO3
4.	The marks of different subject of student Ram is given in the form of	CO3

	dictionary. Now you write a program to find the total marks and the pass percentage scored by Ram. Also sort the marks and the corresponding subjects in descending order. Marks={"m1":92, "m2":56, "m3":85, "m4":78, "m5":82}	
5.	You have developed an online ERP portal app where the students can enter their semester marks of each subject. The app will calculate total marks, percentage and CGP(Cumulative Grade Point) for the semester. However, if a student accidentally inputs an alphabet or non-numeric value instead of a number while entering the marks, the app should display an appropriate message, "Please enter only numbers". As the programmer responsible for developing this app, write a program to handle this exception and ensure that only numeric values are accepted as input. **Note: You can use list to get the 5 subject marks.**	
6.	Mohan created a file called "demo1.txt" where he stored some content. Now you are asked to write a python program to copy all the content of "demo1.txt" file to another file named "demo2.txt " in uppercase. **Example:** **demo1.txt** sathyabama institute of science and technology **demo2.txt** SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY	CO4
7.	Write a python program to add an item in a tuple and after adding the item in tuple, print the tuples in ascending and descending order **Example** **Input :** a=(10,40,50,70,90) Add item=20 **Output:** After adding item : (10,40,50,70,90,20) Ascending order: (10,20,40,50,70,90) Descending order:(90,70,50,40,20,10)	CO3
8.	a) Using proper regular expression get the following output (use sub function) **Sample** **Input :** Ram Suvan work for Suvan Educational Academy **Output :** Ram Buvan work for Buvan Educational Academy b) Write a python program to reverse the content of the file demo1.txt and store it in another file named demo2.txt **Example** **Demo1.txt**: Hello world **Demo2.txt**: dlrow olleH	CO4